

7)Page Frame Replacement Techniques

```
#include <stdbool.h>
#include <stdio.h>

#define MAX_FRAMES 10
#define MAX_PAGES 50

bool isPresent(int frames[], int n, int page) {
    for (int i = 0; i < n; i++)
        if (frames[i] == page)
            return true;
    return false;
}

// Print summary: hits, misses, percentages
void printSummary(const char *algo, int n, int misses) {
    int hits = n - misses;
    printf("\n[%s Summary]\n", algo);
    printf("Total Accesses : %d\n", n);
    printf("Total Hits    : %d\n", hits);
    printf("Total Misses   : %d\n", misses);
    printf("Hit/Miss Ratio : %.2f%%\n", (hits * 100.0) / misses);
}

// FIFO Page Replacement
void fifo(int pages[], int n, int numFrames) {
    int frames[MAX_FRAMES];
    int misses = 0, pointer = 0;

    for (int i = 0; i < numFrames; i++)
        frames[i] = -1;

    printf("\n--- FIFO ---\n");
    printf("%-5s | Frames\n", "Page");

    for (int i = 0; i < n; i++) {
        printf("%-5d | ", pages[i]);

        if (!isPresent(frames, numFrames, pages[i])) {
            frames[pointer] = pages[i];
            pointer = (pointer + 1) % numFrames;
            misses++;
        }
        for (int j = 0; j < numFrames; j++)
```

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        (frames[j] == -1) ? printf("- ") : printf("%d ", frames[j]);
        printf(" *MISS*");
    } else {
        for (int j = 0; j < numFrames; j++)
            (frames[j] == -1) ? printf("- ") : printf("%d ", frames[j]);
        printf(" *HIT*");
    }
    printf("\n");
}
printSummary("FIFO", n, misses);
}

```

// LRU Page Replacement

```

void lru(int pages[], int n, int numFrames) {
    int frames[MAX_FRAMES];
    int lastUsed[MAX_FRAMES];
    int misses = 0;

    for (int i = 0; i < numFrames; i++) {
        frames[i] = -1;
        lastUsed[i] = 0;
    }

    printf("\n--- LRU ---\n");
    printf("%-5s | Frames\n", "Page");

    for (int i = 0; i < n; i++) {
        printf("%-5d | ", pages[i]);

        if (!isPresent(frames, numFrames, pages[i])) {
            int lruIdx = 0;
            for (int j = 1; j < numFrames; j++)
                if (lastUsed[j] < lastUsed[lruIdx])
                    lruIdx = j;

            frames[lruIdx] = pages[i];
            lastUsed[lruIdx] = i + 1;
            misses++;

            for (int j = 0; j < numFrames; j++)
                (frames[j] == -1) ? printf("- ") : printf("%d ", frames[j]);
            printf(" *MISS*");
        } else {
            for (int j = 0; j < numFrames; j++)

```

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        if (frames[j] == pages[i]) {
            lastUsed[j] = i + 1;
            break;
        }

        for (int j = 0; j < numFrames; j++)
            (frames[j] == -1) ? printf("- ") : printf("%d ", frames[j]);
        printf(" *HIT*");
    }
    printf("\n");
}
printSummary("LRU", n, misses);
}

// Optimal Page Replacement
int findOptimalVictim(int frames[], int numFrames, int pages[], int n,
                    int current) {
    int farthest = -1, victimIdx = 0;

    for (int i = 0; i < numFrames; i++) {
        int nextUse = n;
        for (int j = current + 1; j < n; j++) {
            if (pages[j] == frames[i]) {
                nextUse = j;
                break;
            }
        }
        if (nextUse > farthest) {
            farthest = nextUse;
            victimIdx = i;
        }
    }
    return victimIdx;
}

void optimal(int pages[], int n, int numFrames) {
    int frames[MAX_FRAMES];
    int misses = 0, filled = 0;

    for (int i = 0; i < numFrames; i++)
        frames[i] = -1;

    printf("\n--- Optimal ---\n");
    printf("%-5s | Frames\n", "Page");

```

```

for (int i = 0; i < n; i++) {
    printf("%-5d | ", pages[i]);

    if (!isPresent(frames, numFrames, pages[i])) {
        if (filled < numFrames)
            frames[filled++] = pages[i];
        else {
            int victim = findOptimalVictim(frames, numFrames, pages, n, i);
            frames[victim] = pages[i];
        }
        misses++;

        for (int j = 0; j < numFrames; j++)
            (frames[j] == -1) ? printf("- ") : printf("%d ", frames[j]);
        printf(" *MISS*");
    } else {
        for (int j = 0; j < numFrames; j++)
            (frames[j] == -1) ? printf("- ") : printf("%d ", frames[j]);
        printf(" *HIT*");
    }
    printf("\n");
}
printSummary("Optimal", n, misses);
}

int main() {

    printf("Name: ROHAN PODDAR\nUSN:1WA24CS238\n\n");

    int pages[MAX_PAGES], n, numFrames;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter the reference string: ");
    for (int i = 0; i < n; i++)
        scanf("%d", &pages[i]);

    printf("Enter number of frames: ");
    scanf("%d", &numFrames);

    fifo(pages, n, numFrames);
    lru(pages, n, numFrames);
}

```

```
    optimal(pages, n, numFrames);

    return 0;
}
```

Output:

Output

Name: Riya Gupta

USN: 1WA24CS235

Enter number of pages: 12

Enter the reference string: 7

0 1 2 0 3 0 4 2 3 0 3

Enter number of frames: 3

--- FIFO ---

Page	Frames
------	--------

7	7 - - *MISS*
---	--------------

0	7 0 - *MISS*
---	--------------

1	7 0 1 *MISS*
---	--------------

2	2 0 1 *MISS*
---	--------------

0	2 0 1 *HIT*
---	-------------

3	2 3 1 *MISS*
---	--------------

0	2 3 0 *MISS*
---	--------------

4	4 3 0 *MISS*
---	--------------

2	4 2 0 *MISS*
---	--------------

3	4 2 3 *MISS*
---	--------------

0	0 2 3 *MISS*
---	--------------

3	0 2 3 *HIT*
---	-------------

[FIFO Summary]

Total Accesses : 12

Total Hits : 2

Total Misses : 10

Hit/Miss Ratio : 20.00%

Output

--- LRU ---

Page	Frames
7	7 - - *MISS*
0	7 0 - *MISS*
1	7 0 1 *MISS*
2	2 0 1 *MISS*
0	2 0 1 *HIT*
3	2 0 3 *MISS*
0	2 0 3 *HIT*
4	4 0 3 *MISS*
2	4 0 2 *MISS*
3	4 3 2 *MISS*
0	0 3 2 *MISS*
3	0 3 2 *HIT*

[LRU Summary]

Total Accesses : 12
Total Hits : 3
Total Misses : 9
Hit/Miss Ratio : 33.33%

Output

Total Accesses : 12
Total Hits : 3
Total Misses : 9
Hit/Miss Ratio : 33.33%

--- Optimal ---

Page	Frames
7	7 - - *MISS*
0	7 0 - *MISS*
1	7 0 1 *MISS*
2	2 0 1 *MISS*
0	2 0 1 *HIT*
3	2 0 3 *MISS*
0	2 0 3 *HIT*
4	2 4 3 *MISS*
2	2 4 3 *HIT*
3	2 4 3 *HIT*
0	0 4 3 *MISS*
3	0 4 3 *HIT*

[Optimal Summary]

Total Accesses : 12
Total Hits : 5
Total Misses : 7
Hit/Miss Ratio : 71.43%

=== Code Execution Successful ===